

ANGLO-CHINESE JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION

Higher 2

MATHEMATICS

9758/01

Paper 1
QUESTION PAPER

26 August 2025
3 hr

Additional Materials: Printed Answer Booklet
 List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.

Write your answers on the Printed Answer Booklet. Follow the instructions on the front cover of the answer booklet.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are **not** allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You must show all necessary working clearly.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **7** printed pages and **1** blank page.



Anglo-Chinese Junior College

[Turn over

- 1 (a) Without using a calculator, solve the inequality $\frac{x}{x+2} \geq \frac{2}{2-x}$. [4]
- (b) Hence, solve the inequality $\frac{|x|}{|x|+2} \geq \frac{2}{2-|x|}$. [2]
- 2 The graph of $y = \frac{ax^2 + bx + c}{x + d}$ has a vertical asymptote with equation $x = -1$ and an oblique asymptote with equation $y = 2x + 2$. It is given that $c \neq 2$.
- (a) Write down the value of d and show that $a = 2$ and $b = 4$. [3]
- (b) Hence find the range of values of c if the graph has no stationary points. [2]
- 3 It is given that $y = x^{xy}$, where $x > 0$, $y > 0$.
- (a) Show that $\frac{dy}{dx} = \frac{y^2(1 + \ln x)}{1 - \ln y}$. [3]
- (b) Hence find the coordinates of the point on the curve $y = x^{xy}$ whose tangent is parallel to the y -axis. [2]
- 4 The region R is bounded by the curve with equation $y = x \ln x$, the lines $x = e$, $x = e^2$ and the x -axis. Find the exact volume of the solid formed when R is rotated about the x -axis by 2π radians. Give your answer in the form $\frac{\pi e^3}{27}(ae^3 + b)$, where a and b are integers to be found. [6]
- 5 Relative to the origin O , the points A and B have non-zero and non-parallel position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. The plane p has equation $\mathbf{r} \cdot \mathbf{n} = 0$.
- (a) Given that $\mathbf{a} \cdot \mathbf{n} = \mathbf{b} \cdot \mathbf{n} \neq 0$, show that $\overrightarrow{AB}$ is perpendicular to $\mathbf{n}$. Hence, describe the geometrical relationship between $\overrightarrow{AB}$ and the plane p . [2]
- (b) Write down the equation of a line parallel to $\overrightarrow{AB}$ that is contained in the plane p . [1]
- (c) The point F is the foot of perpendicular from a point C with position vector $\mathbf{c}$ to the plane p . Find the position vector of point F , giving your answer in terms of $\mathbf{c}$ and $\mathbf{n}$. [3]

6

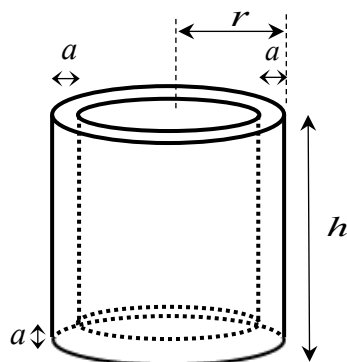


Figure 1

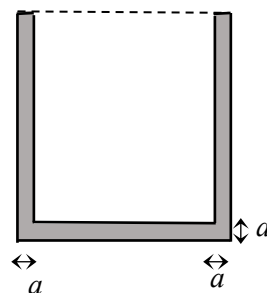


Figure 2

Figure 1 shows an open cylindrical tank. Figure 2 shows the cross-sectional view of the tank. The external radius of the tank is r cm, the external height is h cm and the tank is made with a material of thickness a cm throughout. The internal volume of the tank is fixed at 1000π cm³.

- (a) Show that the volume of the material needed to make the cylindrical tank is given by

$$V = k\pi \left[\frac{r^2}{(r-a)^2} - 1 \right] + \pi r^2 a, \text{ where } k \text{ is a constant to be determined.} \quad [3]$$

- (b) Find, in terms of a , the value of r that minimises V . (You need not show that your answer gives a minimum.) [3]

- 7 A freshly brewed cup of tea, initially at a temperature of 85°C , is left out to cool in a room with a room temperature of 25°C . After 20 minutes, the temperature of the tea is 55°C .

It is known that the rate of change of the temperature of the tea is proportional to the difference between the temperature of the tea and the room temperature.

- (a) Given that θ is the temperature of the tea t minutes after the tea is left out to cool, show that $\theta = 25 + 60e^{-kt}$, where k is a constant to be determined. [5]

- (b) Sketch the graph of θ against t . [2]

- 8 The curve with equation $y = f(x)$ for $x \leq b$, where $f(x)$ is a **quadratic** function, intersects the x -axis at the points $(a, 0)$ and $(b, 0)$, and the y -axis at the point $(0, c)$. The graph of $y = f(x)$ is shown in Figure 1. The scales on the x - and y -axes are the same.

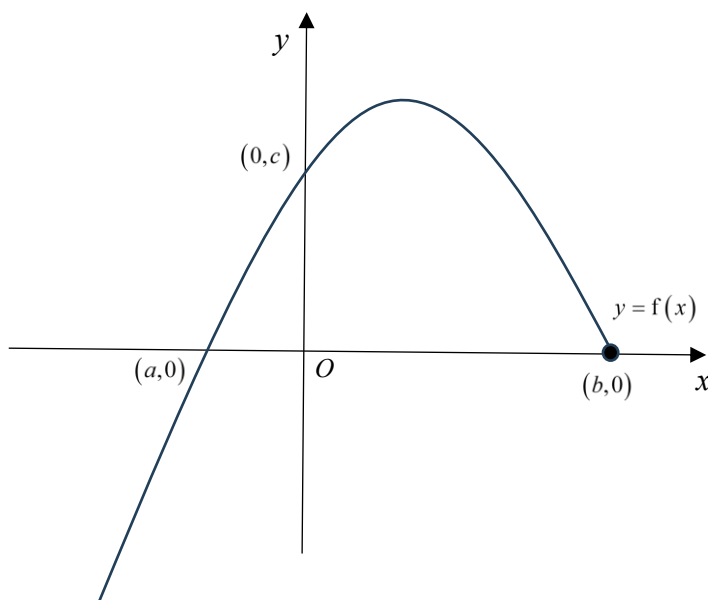


Figure 1

- (a) (i) Sketch the graph of $y = f(|x|)$, labelling the points where the graph intersects or touches the axes. [2]
- (ii) Sketch the graph of $y = \frac{1}{f(x)}$, labelling the points where the graph intersects or touches the axes, as well as the equations of any asymptotes. [2]
- (iii) Describe fully a sequence of transformations which transforms the graph of $y = f(2x+1)$ onto the graph of $y = f(x)$. [2]
- (b) The function $y = g(x)$ is such that $g(x) = f(x)$ for $x \leq k$, and $y = g^{-1}(x)$ exists.
- (i) Find the largest possible value of k in terms of a and b . [1]
- (ii) Using the value of k found in (b)(i), sketch the graph of $y = g^{-1}(x)$ on Figure 1 in the Answer Booklet, labelling the intersections with the axes. Write down the range of g^{-1} in terms of a and b . [2]
- (iii) Explain why the solution to $g^{-1}(x) = x$ satisfies the equation $g^{-1}(x) = g(x)$. [1]

- 9 The planes π_1 and π_2 have equations $3x + c(y + z) - 2 = 0$ and $\mathbf{r} = (\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}) + s(2\mathbf{i} - \mathbf{j} + 3\mathbf{k}) + t(\mathbf{i} - \mathbf{k})$ respectively, where c is a constant, and s and t are parameters. The point $A(1, 3, -2)$ lies in both planes.

(a) Show that $c = -1$. [1]

(b) Show that the vector equation of the line of intersection of π_1 and π_2 , line l , is given by $\mathbf{r} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k} + \alpha(\mathbf{i} - \mathbf{j} + 4\mathbf{k})$, where α is a parameter. [3]

(c) Find the position vectors of the points on the line l which are a distance of $3\sqrt{2}$ from the point $B(2, -3, 7)$. [4]

(d) Find the equation of the plane π_3 which is parallel to π_2 and contains the point B .

Hence show that the distance between the planes π_2 and π_3 is $\frac{20}{3\sqrt{3}}$. [3]

10 Do not use a calculator in answering this question.

(a) One of the roots of the equation $\omega^4 - 2\omega^3 + 10\omega^2 + p\omega + q = 0$, where p and q are real, is $2 + 3i$. Find the values of p and q and the other roots of the equation. [6]

(b) The complex numbers u and v are such that $u = -\sqrt{2} + i\sqrt{2}$, and $|v| = 3$ and $\arg v = \theta$, where $0 < \theta < \frac{\pi}{4}$. The points A , B and C represents u , v and $u + v$ respectively on an Argand diagram.

(i) Find the modulus and argument of u . [2]

(ii) Sketch the points A , B and C on an Argand diagram. [2]

(iii) By finding the angle OAC in terms of θ or otherwise, show that

$$|u + v|^2 = a + b \cos(\theta + K), \text{ where } a, b \text{ and } K \text{ are constants to be determined. [3]}$$

- 11** The curve C is defined by the parametric equations $x = at^2$, $y = \frac{a}{t}$, $t \neq 0$, where a is a positive constant.

(a) Show that the equation of the tangent to the curve at the point $P\left(ap^2, \frac{a}{p}\right)$ is

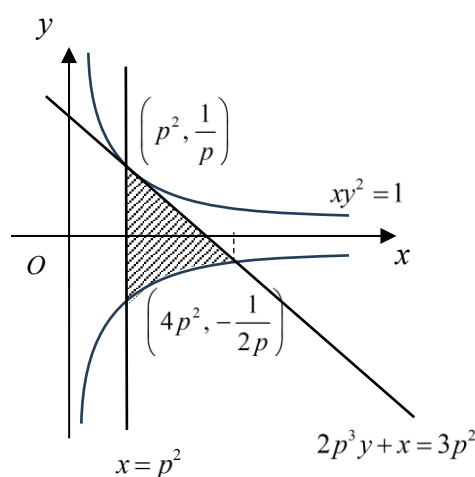
$$2p^3y + x = 3ap^2. \quad [2]$$

(b) Find the equations of the tangents to C at the points where $x = a$. Find also the acute angle between these tangents. [3]

(c) (i) Show that $(q - p)^2(q + 2p) = q^3 - 3p^2q + 2p^3$. [1]

(ii) The tangent to the curve at P cuts the curve again at $Q\left(aq^2, \frac{a}{q}\right)$. Find q in terms of p . [3]

The following diagram shows the graph of C for $a = 1$, and the Cartesian equation of the curve C is $xy^2 = 1$. The tangent to C at the point $\left(p^2, \frac{1}{p}\right)$ cuts the curve again at the point $\left(4p^2, -\frac{1}{2p}\right)$.



The region S is bounded by the tangent to C at the point $\left(p^2, \frac{1}{p}\right)$, the curve C and the line $x = p^2$.

(d) Find the area of S in terms of p . [3]

- 12** Sarah, a Singaporean Citizen, started a 4-year undergraduate Computing course at the National University of Singapore (NUS) in August 2021. The annual tuition fee, after the MOE Tuition Grant, is \$8,250. Sarah took a Tuition Fee Loan (TFL) covering 90% of her tuition fees. This loan is interest-free during her course of study.

Sarah decided to set aside \$ x per month from August 2021. This amount is increased by \$2 each subsequent month. (Sarah set aside \$ x in August 2021, \$ $(x + 2)$ in September 2021,) With the amount set aside, she repaid one-fifth of her total TFL at the end of July 2025.

- (a)** Find the value of x . [3]

Upon Sarah's graduation, the TFL's interest rate becomes effective at a fixed rate of 0.4% per month, calculated on the outstanding balance at the end of the month. The interest is first charged on 31st August 2025.

Sarah plans to make monthly repayments of \$ y to the bank on the 1st of every month, starting from 1st September 2025.

- (b)** Find the amount of outstanding loan after Sarah's first monthly repayment. [1]

- (c)** Show that the amount of outstanding loan after Sarah's n th monthly repayment is $23760(1.004)^n - 250y(1.004^n - 1)$. [3]

- (d)** Sarah wants to pay off her remaining loan by the end of 3 years. Find the value of y . [2]

Sarah decided on a different repayment strategy. For the first 24 months, she repays \$500 per month. After these 24 months, she doubles her monthly repayment to \$1000.

- (e)** Determine the month and year Sarah will clear her loan and calculate the amount of her final repayment. [4]

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